

Mean-value of the Riemann zeta-function on the critical line

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Abstract. This is an expository article. It is a collection of some important results on the mean-value of $|\zeta(\frac{1}{2} + it)|$.

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1. Introduction

This article is a collection of some important results on the mean value of $|\zeta(\frac{1}{2} + it)|$ and its powers. Section 2 deals with the mean square and §3 with the mean fourth power and the mean twelfth power. Section 4 deals with the general $(2k)$ th power where $k > 0$ and §5 gives other important results.

2. Mean square

In 1918 Hardy and Littlewood [12] proved that

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^2 dt \sim T \log T.$$

In 1926 this was improved by Ingham [18] to

$$\frac{1}{2\pi} \int_0^T |\zeta(\tfrac{1}{2} + it)|^2 dt = \frac{T}{2\pi} \log \frac{T}{2\pi} + (2\gamma - 1) \frac{T}{2\pi} + E(T),$$

where $E(T) = O(T^{1/2} \log T)$. More complicated argument due to Titchmarsh [33] showed that $E(T) = O(T^{5/12} (\log T)^2)$. The final improvement of this theorem was published in 1978 by Balasubramanian [3] who proved that $E(T) = O(T^{1/3})$, $E(T) = O(T^{27/82})$, and so on. It should be mentioned that subsequent to this discovery Good [10] gave a variant of Balasubramanian's proof. It should also be mentioned that Heath-Brown's powerful method [14] which gives the mean fourth power with error term also gives the theorem of Balasubramanian.

Titchmarsh [34, 35] had introduced a new mean value. He had proved that, as $\delta \rightarrow 0$

$$\int_0^\infty |\zeta(\tfrac{1}{2} + it)|^2 \exp(-\delta t) dt \sim \frac{1}{\delta} \log \frac{1}{\delta}$$

and

$$\int_0^\infty |\zeta(\tfrac{1}{2} + it)|^4 \exp(-\delta t) dt \sim \frac{1}{2\pi^2 \delta} \left(\log \frac{1}{\delta} \right)^4.$$

Kober [22] proved that

$$\int_0^\infty |\zeta(\tfrac{1}{2} + it)|^2 \exp(-2\delta t) dt = \frac{\gamma - \log(4\pi\delta)}{2 \sin \delta} + \sum_{n=0}^N c_n \delta^n + O(\delta^{N+1}),$$

where the 0-constant depends on N . A simpler proof was given by Atkinson [1]. Regarding Ω theorems for $E(T)$ Good [11] developed a method to prove that $E(T) = \Omega(T^{1/4})$. Heath-Brown [15] developed a more powerful method which actually gives

$$\int_0^T (E(t))^2 dt = cT^{3/2} + O(T^{5/4} (\log T)^2),$$

where $c = \frac{2}{3}(2\pi)^{-1/2}(\zeta(\frac{3}{2}))^4(\zeta(3))^{-1}$.

3. Mean fourth power

In 1922 Hardy and Littlewood [13] proved that

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^4 dt = O(T(\log T)^4).$$

This was improved in 1926 by Ingham [18] to

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^4 dt = \frac{T}{2\pi^2} (\log T)^4 + O(T (\log T)^3)$$

by using the approximate functional equation for $(\zeta(s))^2$. A simpler proof of this fact was given recently by Ramachandra [24] using the theorem of Montgomery and Vaughan. Subsequently Heath-Brown [14] showed by deep arguments that

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^4 dt = TP(\log T) + O(T^{\frac{7}{8}+\epsilon}),$$

where $P(x)$ is a polynomial of degree 4 in x . The coefficient of x^4 is $1/2\pi^2$ and that of x^3 is

$$2\pi^{-2}[4\gamma - 1 - \log(2\pi) - 12\pi^{-2}\zeta'(2)].$$

He also proved another deep result [16] which asserts that

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^{12} dt = O(T^2 (\log T)^{17}).$$

Previously Atkinson [2] had proved that

$$\int_0^\infty |\zeta(\tfrac{1}{2} + it)|^4 \exp(-\delta t) dt = \frac{1}{\delta} Q\left(\log \frac{1}{\delta}\right) + O\left(\left(\frac{1}{\delta}\right)^{\frac{3}{8}+\epsilon}\right)$$

where $Q(x)$ is a polynomial of degree 4 in x . The coefficient of x^4 is $1/2\pi^2$ and that of x^3 is

$$-\frac{1}{\pi^2} \left[2 \log(2\pi) - 6\gamma + \frac{24\zeta'(2)}{\pi^2} \right].$$

For the fourth power mean Iwaniec [19] proved the following deep result. Let I_r ($r = 1, 2, \dots, R$) be R disjoint intervals of length T_0 satisfying $T^{1/2} \leq T_0 \leq T$. Then

$$\sum_{r=1}^R \int_{I_r} |\zeta(\tfrac{1}{2} + it)|^4 dt \ll_\varepsilon (RT_0 + T(R/T_0)^{1/2}) T^\varepsilon.$$

Choosing $R = 1$, $T_0 = T^{2/3}$ we obtain

$$\int_T^{T+T^{2/3}} |\zeta(\tfrac{1}{2} + it)|^4 dt \ll T^{3+\varepsilon}.$$

The deep result of Iwaniec also implies

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^{12} dt \ll T^{2+\varepsilon}.$$

This can be seen as follows. It suffices to prove that

$$\int_T^{2T} |\zeta(\tfrac{1}{2} + it)|^{12} dt \ll T^{2+\varepsilon}.$$

The contributions to the twelfth power moment from those t with $|\zeta(\tfrac{1}{2} + it)| \leq T^{\frac{1}{6}+\varepsilon}$ are together $\ll T^{2+9\varepsilon}$ using $\int_0^T |\zeta(\tfrac{1}{2} + it)|^4 dt \ll T(\log T)^4$. So one has to consider the range

$$T^{\frac{1}{6}+\varepsilon} \leq |\zeta(\tfrac{1}{2} + it)| \leq T^{\frac{1}{6}+\varepsilon}$$

since $\mu(\tfrac{1}{2}) \leq \tfrac{1}{6}$. We divide the range $[T^{\frac{1}{6}+\varepsilon}, T^{\frac{1}{6}+\varepsilon}]$ into $[T^{\frac{1}{6}+\varepsilon}, 2T^{\frac{1}{6}+\varepsilon})$, $[2T^{\frac{1}{6}+\varepsilon}, 4T^{\frac{1}{6}+\varepsilon})$, $[4T^{\frac{1}{6}+\varepsilon}, 8T^{\frac{1}{6}+\varepsilon})$, ... the last interval either terminates at $T^{\frac{1}{6}+\varepsilon}$ or goes just a little beyond. In any typical interval $[V, 2V)$ we have $V \leq |\zeta(\tfrac{1}{2} + it)| < 2V$. For any given V we consider the set $G(V)$ of all t with $V \leq |\zeta(\tfrac{1}{2} + it)| < 2V$. We choose a small $\delta > 0$ to be fixed later. We next divide $[T, 2T]$ into disjoint intervals $[T, T+V^{4-\delta})$, $[T+V^{4-\delta}, T+2V^{4-\delta})$, $[T+2V^{4-\delta}, T+3V^{4-\delta})$, ... the last interval either terminating at $2T$ or going just a little beyond $2T$. Out of these we retain only those which have at least one point of $G(V)$. With every such interval $[A, B)$ we associate the interval I : $(A - (\log T)^2, B + (\log T)^2)$. The intervals I are not disjoint. But every point of $[A, B)$ is interior to at most two intervals I . Hence if we put $T_0 = V^{4-\delta} + 2(\log T)^2$ we have

$$\sum_I \int_I |\zeta(\tfrac{1}{2} + it)|^4 dt \ll (RV^{4-\delta} + T(R/V^{4-\delta})^{1/2}) T^\varepsilon.$$

By convexity each of the integrals on the left $\geq V^4 (\log T)^{-2}$. Hence

$$RV^4 (\log T)^{-2} \ll (RV^{4-\delta} + T(R/V^{4-\delta})^{1/2}) T^\varepsilon.$$

We now put $\delta = 16\varepsilon$. It follows that

$$RV^4 (\log T)^{-2} \ll (R/V^{4-\delta})^{1/2} T^{1+\varepsilon}.$$

Hence $R \ll \frac{T^{2+\varepsilon}}{V^{12-\delta}} \ll \frac{T^{2+100\varepsilon}}{V^{12}}$. On the other hand by Iwaniec's result we have also

$$\sum_I \int_I |\zeta(\tfrac{1}{2} + it)|^4 dt \ll_\varepsilon (RT_0 + T(R/T_0)^{1/2}) T^\varepsilon.$$

Here the first term on the right side can be ignored and the left side is

$$\geq \sum_I \mu(I) V^4$$

where $\mu(I)$ is the measure of $G(V) \cap I$. Hence

$$\sum_I \mu(I) \ll \frac{1}{V^4} T^{1+\varepsilon} (R/T_0)^{1/2}$$

and so

$$\sum_I \mu(I) V^{12} \ll V^8 T^{1+\varepsilon} (T^{2+100\varepsilon}/V^{16-\delta})^{1/2} \ll T^{2+100\varepsilon}$$

i.e.

$$\int_{G(V)} |\zeta(\tfrac{1}{2} + it)|^{12} dt \ll T^{2+100\varepsilon}$$

Summing up over all V the twelfth power moment stated in the beginning is proved.

4. Mean $(2k)$ th power

Titchmarsh [36] was the first to prove the following result. For every fixed positive integer k ,

$$\int_0^\infty |\zeta(\tfrac{1}{2} + it)|^{2k} \exp(-\delta t) dt \gg_k \frac{1}{\delta} \left(\log \frac{1}{\delta} \right)^{k^2}.$$

This gives

$$\frac{1}{T} \int_T^{2T} |\zeta(\tfrac{1}{2} + it)|^{2k} dt = \Omega((\log T)^{k^2}).$$

Put

$$M(k, T, T+H) = \frac{1}{H} \int_T^{T+H} |\zeta(\tfrac{1}{2} + it)|^{2k} dt$$

where k is a positive real number and

$$T \geq H \geq 100 \log \log T \geq C_0$$

where C_0 is a large positive constant. Ramachandra [25] was the first to prove that if $2k$ is a positive integer then $M(k, T, T+H) \gg_k (\log H)^{k^2}$. This gives

$$\frac{1}{T} \int_T^{2T} |\zeta(\tfrac{1}{2} + it)|^{2k} dt \gg_k (\log T)^{k^2}.$$

Using Gabriel's convexity theorem this was upheld by Heath-Brown [17] for all rational $k > 0$. Regarding upper bounds Ramachandra [26] was the first to prove that

$$\frac{1}{T} \int_T^{2T} |\zeta(\tfrac{1}{2} + it)| dt \ll (\log T)^{1/4}.$$

In fact he [26] proved that $M(\tfrac{1}{2}, T, T+T^{1/2+\varepsilon}) \ll (\log T)^{1/4}$ without any hypothesis and $M(\tfrac{1}{2}, T, T+T^{1/4+\varepsilon}) \ll (\log T)^{1/4}$ on Riemann hypothesis. Heath-Brown [17] proved

again by using Gabriel's convexity theorem that $M\left(\frac{1}{m}, T, 2T\right) \ll (\log T)^{1/m^2}$ where $m \geq 1$ is an integer. Jutila [21] proved that

$$(\log T)^{1/m^2} \ll M\left(\frac{1}{m}, T, 2T\right) \ll (\log T)^{1/m^2}$$

where the constants implied by the Vinogradov symbol $\ll$ are independent of m . He applied this to show that the measure of the set $|\zeta(\frac{1}{2} + it)| \geq V$ is

$$\leq \exp\left(-\frac{(\log V)^2}{\log \log T} \left(1 + O\left(\frac{\log V}{\log \log T}\right)\right)\right)$$

where $1 \leq V \leq \log T$.

Regarding lower bounds Ramachandra [27] was the first to prove

$$M(k, T, T+H) \gg (\log H)^{k^2} (\log \log H)^{-C}$$

where $k > 0$ is any real constant and C depends only on k . Using Heath-Brown's idea he [28] proved

$$M(k, T, T+H) \gg_k (\log H)^{k^2}$$

where $k > 0$ is any rational constant. Using the same idea he proved [29, 30] that for irrational k

$$M(k, T, T+H) \gg g\left(\frac{\log H}{\log \log H}\right)^{k^2},$$

where g is a function of H which tends to infinity with H .

Ramachandra [31] has proved (see also Balasubramanian and Ramachandra [4]) that if k is an integer subject to $1 \leq k \leq \log H$, we have uniformly in k, T, H

$$M(k, T, T+H) \gg \sum_{n \leq H/100} \frac{(d_k(n))^2}{n} \left(1 - \frac{\log n}{\log H} + \frac{1}{\log \log H}\right).$$

Put $\Lambda = (\text{RHS})^{1/2k}$. The quantity Λ has been studied as a function of k by Balasubramanian. It was shown by Balasubramanian and Ramachandra [5] that the maximum of Λ as k varies is attained in $1 \leq k \leq \log H$ and that $\log \max \Lambda \gg \left(\frac{\log H}{\log \log H}\right)^{1/2}$. By an ingenious argument Balasubramanian [6] showed that

$$\log \max \Lambda \sim \alpha \left(\frac{\log H}{\log \log H}\right)^{1/2},$$

where $\alpha = \frac{1}{2} \max_{l > 0, l \text{ is real}} \{(\exp(-2l) + \int_{2l}^{\infty} \frac{e^{-\theta}}{\theta} d\theta)(2l \exp(2l)^{1/2})\}$.

Richert showed by his pocket calculator that $\alpha = 0.75$ Thus

$$\max_{T \leq t \leq T+H} |\zeta(\frac{1}{2} + it)| > \exp\left(\frac{3}{4} \left(\frac{\log H}{\log \log H}\right)^{1/2}\right)$$

where $T \geq H \geq 100 \log \log T$ and $T > C_0$. For the earlier history of the problem see [32] and [5]. This is an improvement on the result

$$\max_{0 \leq t \leq T} |\zeta(\tfrac{1}{2} + it)| > \exp\left(\frac{1}{20} \left(\frac{\log T}{\log \log T}\right)^{1/2}\right)$$

for $T \geq T_0$ due to Montgomery [23] who proved this on the assumption of the Riemann hypothesis.

5. Other important results

Let

$$M(s) = \sum_{m \leq M} a_m m^{-s}$$

and

$$S(T, M) = \int_0^T |\zeta(\tfrac{1}{2} + it) M(it)|^2 dt.$$

Iwaniec [21] was the first to prove that

$$S(T, M) \ll T^{1+\varepsilon} \sum_{m \leq M} |a_m|^2$$

for $M \leq T^{1/2}$ unconditionally and for $M \leq T^{4/7}$ conditionally. In the case $a_m = \mu(m)$ Heath-Brown has proved this result unconditionally for $M \leq T^{8/15}$ (see p. 306 of [8]). In a series of papers Deshouillers and Iwaniec [8, 9] have proved this unconditionally when $a_m = \Lambda(m)$ or $a_m = \mu(m)$ for $M \leq T^{4/7}$.

Iwaniec [20] was the first to prove that

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^4 |M(it)|^2 dt \ll T^{1+\varepsilon} \sum_{m \leq M} |a_m|^2$$

for $M \leq T^{1/10}$. Deshouillers and Iwaniec prove this for $M \leq T^{1/5}$. Asymptotic formulae are also important (*e.g.* for the discussion of the zeros on the critical line). In this connection Balasubramanian and Conrey [7] have proved that

$$\int_0^T |\zeta(\tfrac{1}{2} + it) M(\tfrac{1}{2} + it)|^2 dt \sim T \sum_{m_1, m_2 \leq M} \frac{a_{m_1} a_{m_2}}{[m_1, m_2]} \left(\log \frac{T(m_1, m_2)}{2\pi m_1 m_2} + 2\gamma - 1 \right)$$

Provided $a_m = \mu(m)f(m)$ where f is a smooth function $\ll 1$ and $M \leq T^{\frac{8}{13}-\varepsilon}$. Using this and refining the method of Levinson they proved that 38% of the zeros lie on the critical line.

In the theory of the Riemann zeta-function an indispensable book is [37]. This has been very useful in the preparation of this article.

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